MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

RELAY, 4PDT, ANALOG, KEY CODE NRN

This amendment forms a part of Military Specification MIL-M-28787/133B(EC), dated 10 April 1979, and is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 1.2, Contact current: Delete "10 A" and substitute "13 A".

PAGE 3

3.5.8, second line: Delete "M28787/133" and substitute "M28787/133-1".

PAGE 5

* 3.10, item a: Delete "10 A" and substitute "13 A".

PAGES 5 AND 6

4.3, delete and substitute new paragraph as follows:

"4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film unless otherwise specified."

PAGE 7

TABLE I, Test conditions D1-D61, delete and substitute the following:

D1-D60	Pin jumper continuity	V	S	0.0 (0.0)	25 (75)	0.0 (0.0)	25 (75)	0.0 (0.0)	25 (75)	0.0 (0.0)	25 (75)	mΩ (mV)
D61	Pin jumper continuity	V	S	0.0 (0.0)	1.0 (3.0)	0.0 (0.0)	1.0 (3.0)	0.0 (0.0)	1.0 (3.0)	0.0 (0.0)	1.0 (3.0)	Ω (V)

PAGE 9

* 4.5.5.1, second line: Delete "10 $^{+1.0}_{-0.2}$" and substitute "13 $^{+1.0}_{-0.2}$".

PAGE 10

* Change "6.3 Changes from previous issue" to "6.4 Changes from previous issue". Add new paragraph 6.3 as follows.

"6.3 Cross reference of classification. Military Specification MIL-M-28787/133 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2658674. Part number M28787/133-1 supersedes part numbers 2658674 and M28787/133."

PAGE 14

FIGURE 3: Delete and substitute new figure 3 as printed on page 3 of this amendment.

FSC 5963
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PAGE 15

* TABLE IV: Delete and substitute new table IV as printed on page 4 of this amendment.

PAGE 16

* FIGURE 4: Delete and substitute new figure 4 as printed on page 5 of this amendment.

PAGE 20

TABLE V, Test condition D61: Max column, delete "25 (75)" and substitute "1.0 (3.0)"; unit column, delete "mΩ (mV)" and substitute "Ω (V)".

PAGE 22

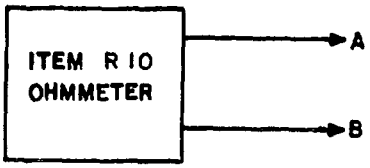
* FIGURE 5: Delete and substitute new figure 5 as printed on page 6 of this amendment.

PAGE 24

* FIGURE 6: Delete and substitute new figure 6 as printed on page 7 of this amendment.

NOTE: The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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NRN MODULE

1A			1B
2			2
3			3
4			4
22	A (NO)	A (NC)	22
23			23
24			24
5			5
6			6
7			7
25	A (COM)	B (NO)	25
26			26
27			27
8			8
9			9
10			10
28	B (NC)	B (COM)	28
29			29
30			30
12			15
13			16
14			17
32	C (NO)	C (COM)	35
33			36
34			37
18			11
19			12
20			13
38	D (NO)	C (NC)	31
39			32
40			33
15			18
16			19
17			20
35	D (COM)	D (NC)	38
36			39
37			40

21 (R3)

1 (+Vdc) (DIODE RET) 1

31 (10 Vdc RET) 25 Vdc RET (<10 ms) 21

25 Vdc RET (≥10 ms) 14

11 (FRAME) 34 Vdc RET (<10 ms) 34

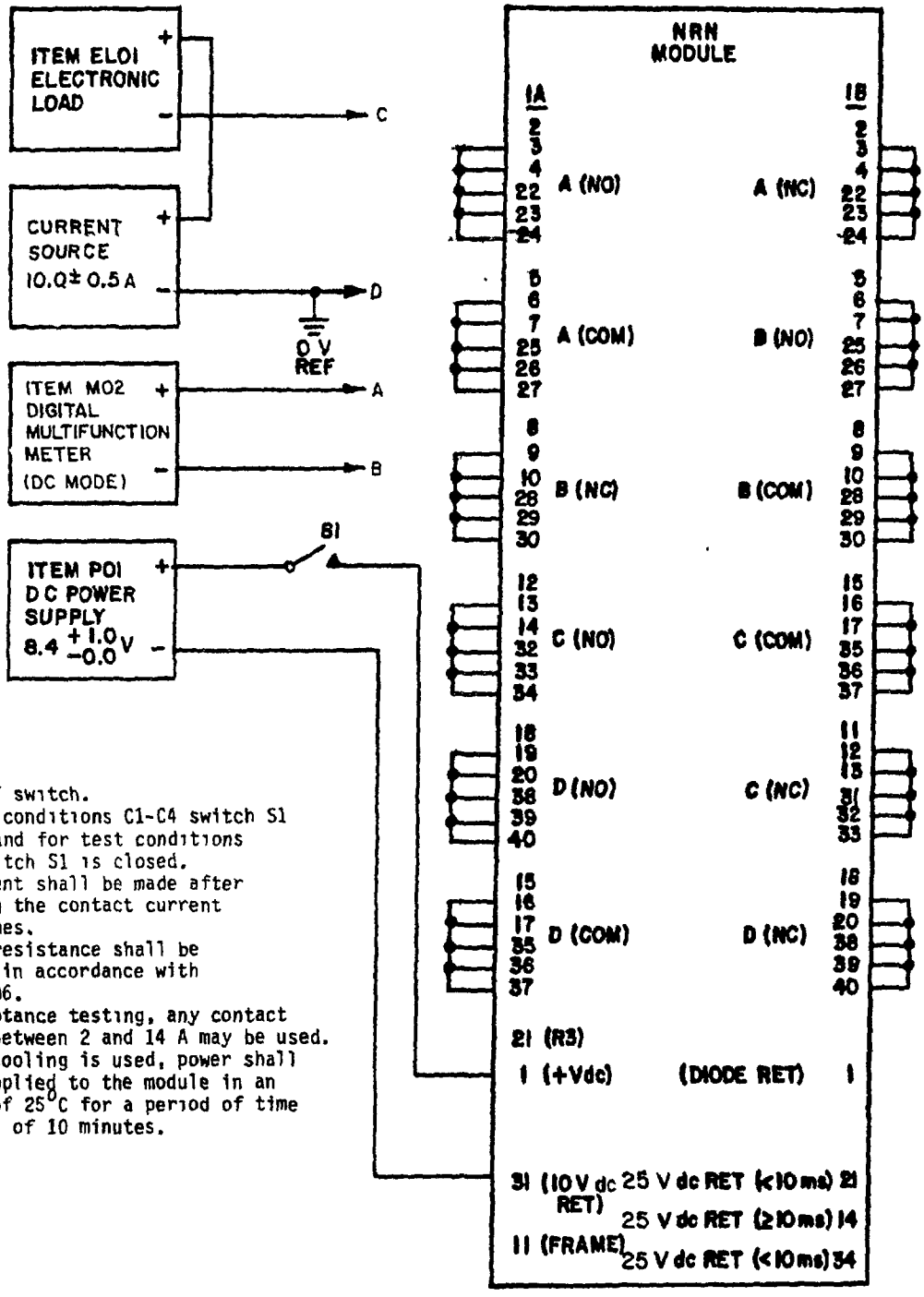
FIGURE 3. Coil resistance and resistor test circuit for tables II and III.

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TABLE IV. Contact resistance test.

Test condition	Parameter	Connect				Initial limits 25°C ±5° ambient		Unit	Test Figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
C1	NC Contact resistance	A5	B2	A27	B24	0.0	16.0	mV/A	4
C2	NC "	A8	A8	B30	A30	"	"	"	"
C3	NC "	B15	B11	B37	B33	"	"	"	"
C4	NC "	A15	B18	A37	B40	"	"	"	"
C5	NO "	A2	A5	A24	A27	"	"	"	"
C6	NO "	B5	B8	B27	B30	"	"	"	"
C7	NO "	A12	B15	A34	B37	"	"	"	"
C8	NO "	A18	A15	A40	A37	"	"	"	"

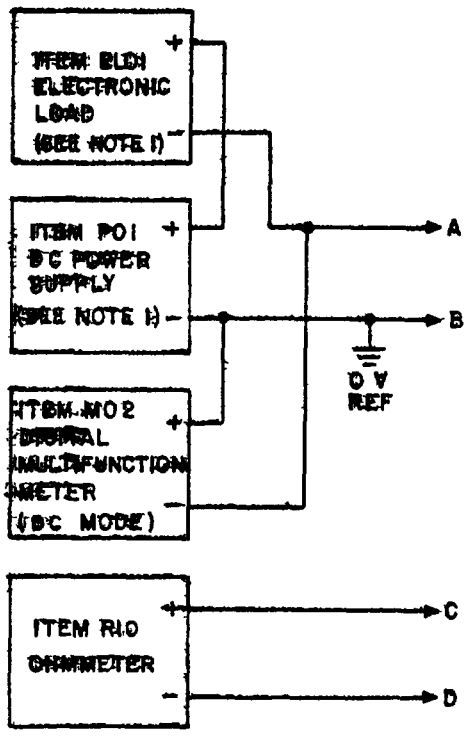
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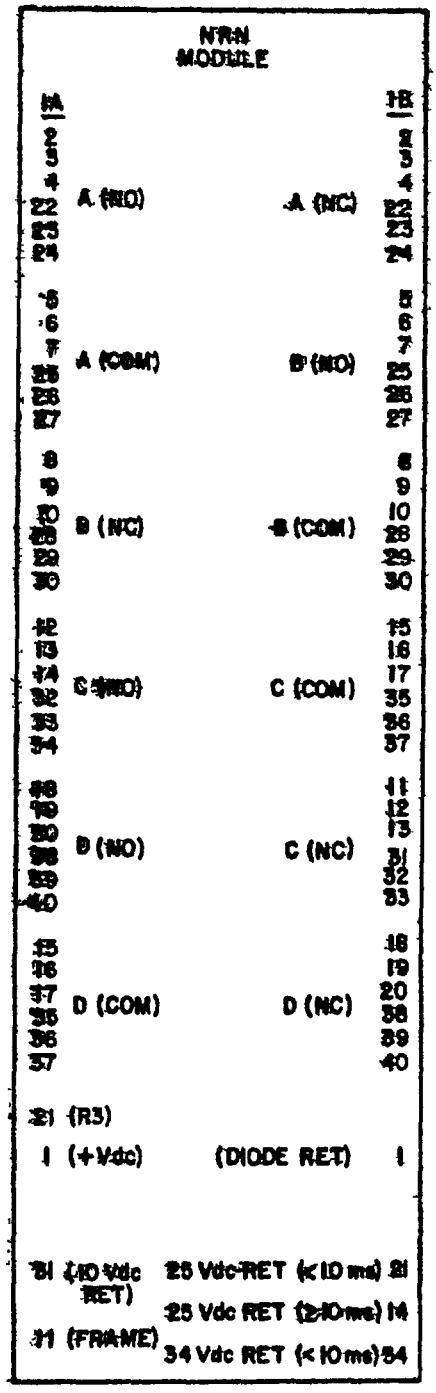


NOTES:
1. S1 = SPST switch.
2. For test conditions C1-C4 switch S1 is open and for test conditions C5-C8 switch S1 is closed.
3. Measurement shall be made after switching the contact current three times.
4. Contact resistance shall be measured in accordance with MIL-R-6106.
5. For acceptance testing, any contact current between 2 and 14 A may be used.
6. When no cooling is used, power shall not be applied to the module in an ambient of 25°C for a period of time in excess of 10 minutes.

FIGURE 4. Contact resistance test circuit for table IV.

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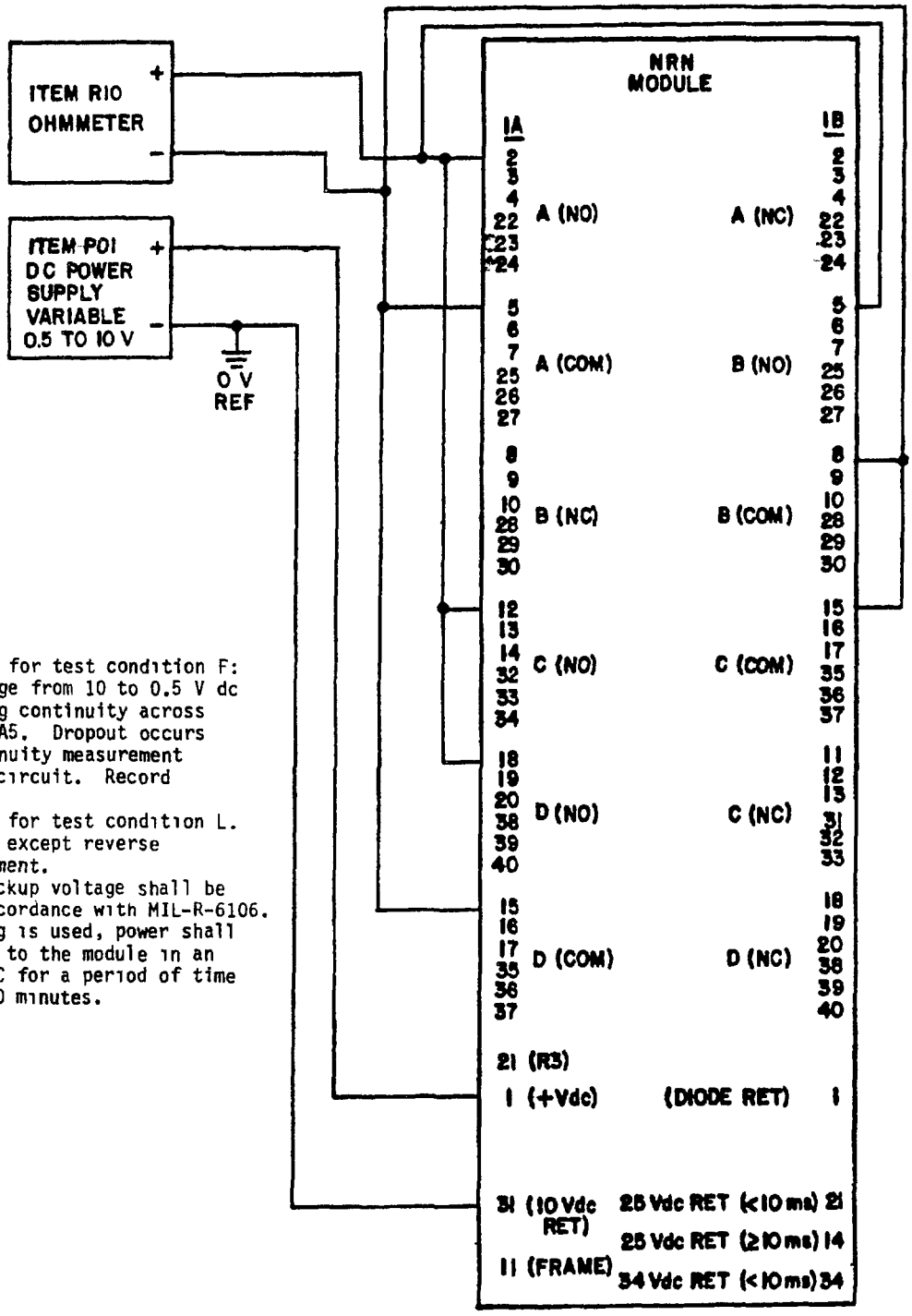




NOTES:
1. Adjust item P06 for 10.0 V ±10% then adjust item EL01 for 3.00 ±0.03 A at the module pin.
2. When no cooling is used, power shall not be applied to the module in an ambient of 25°C for a period of time in excess of 10 minutes.
3. Test conditions E1 and E2 shall be measured using a current of 1 mA maximum.

FIGURE 5. Pin jumper continuity and diode resistance test circuit for tables V and VI.

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NOTES:
1. Test procedure for test condition F: Decrease voltage from 10 to 0.5 V dc while observing continuity across pins 1A2 and 1A5. Dropout occurs when the continuity measurement shows an open circuit. Record voltage level.
2. Test procedure for test condition L. Same as note 1 except reverse voltage adjustment.
3. Dropout and pickup voltage shall be measured in accordance with MIL-R-6106.
4. When no cooling is used, power shall not be applied to the module in an ambient of 25°C for a period of time in excess of 10 minutes.

FIGURE 6. Dropout and pickup voltage test circuit for tables VII and XII.

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Preparing activity:
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(Project 5963-N568)

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